

Econometrics II – Assignment 3: A Dynamic Model for U.S. House Prices

Spring 2026

1 Introduction

Housing is the largest asset on the balance sheet of most U.S. households and house-price swings have, historically, propagated to the wider macroeconomy; the user-cost framework predicts that, in equilibrium, prices are tied to rents and to the mortgage rate. This report tests that prediction by estimating the long-run relation

$$h_t = \alpha + \beta_1 r_t + \beta_2 m_t, \quad (1)$$

where h_t is log house prices, r_t is log rent, and m_t is the 30-year fixed mortgage rate. Using monthly data for 2012M1–2023M12, we estimate a CVAR with $k = 4$ lags and an unrestricted constant by Gaussian maximum likelihood, test the strict user-cost restriction $\beta_1 = 1$ by a Johansen LR test, and check the sensitivity of the conclusion to extending the sample back to 2000M1.

We find one co-integrating relation, but a rent elasticity statistically above one and a mortgage coefficient indistinguishable from zero, so the strict user-cost benchmark is rejected; equilibrium adjustment is borne almost entirely by the mortgage rate, and the finding is sensitive to the inclusion of the 2008-09 crisis years.

2 Description of the Data

The data come from `Assignment_3.xlsx` and are based on FRED series for the S&P/Case-Shiller U.S. National Home Price Index, the CPI for rent of primary residence, and the 30-year fixed mortgage rate. The transformed variables are $h_t = \log(\text{HousePrice})$, $r_t = \log(\text{Rent})$, and $m_t = \text{MortgageRate}$. The effective sample contains 144 monthly observations.

Figure 1 shows that both h_t and r_t trend upward, while m_t is much more cyclical and rises sharply in 2022–2023. The visual pattern is consistent with non-stationary behaviour and motivates a co-integration analysis rather than a stationary VAR in levels. Table 1 reports basic moments of the three series.

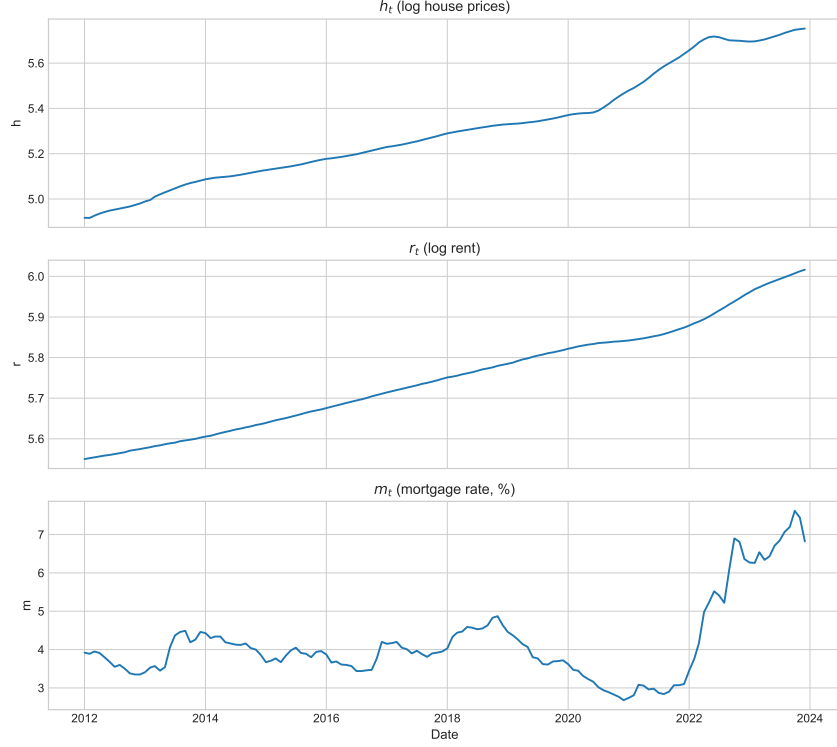


Figure 1: Monthly series for h_t , r_t , and m_t , 2012M1–2023M12.

Table 1: Summary statistics for the analysis variables, 2012M1–2023M12.

Variable	Mean	Std. Dev.	Min	Max
h	5.3108	0.2402	4.9165	5.7527
r	5.7511	0.1287	5.5501	6.0165
m	4.1642	1.0691	2.6800	7.6200

3 Econometric Theory

ADF testing. For each variable y_t we run an augmented Dickey–Fuller regression $\Delta y_t = (\rho - 1)y_{t-1} + \sum_{j=1}^p \phi_j \Delta y_{t-j} + c + \varepsilon_t$ and test $H_0 : \rho = 1$ against $\rho < 1$. The statistic is the t -ratio on $(\rho - 1)$ and its asymptotic distribution is the non-standard Dickey–Fuller distribution (not $N(0, 1)$); we use the constant-only critical values throughout. Lag length is selected by AIC.

The CVAR. Let $X_t = (h_t, r_t, m_t)'$. Since the three series are treated as $I(1)$, the natural framework is a co-integrated vector autoregression in error-correction form,

$$\Delta X_t = \Pi X_{t-1} + \Gamma_1 \Delta X_{t-1} + \cdots + \Gamma_{k-1} \Delta X_{t-k+1} + \mu + e_t, \quad (2)$$

where Π contains long-run information, Γ_j capture short-run dynamics, and e_t is a zero-mean innovation. If Π has reduced rank r , we can factorize $\Pi = \alpha\beta'$, so

$$\Delta X_t = \alpha\beta' X_{t-1} + \Gamma_1 \Delta X_{t-1} + \cdots + \Gamma_{k-1} \Delta X_{t-k+1} + \mu + e_t. \quad (3)$$

Here β contains the co-integrating vectors and $\alpha_h, \alpha_r, \alpha_m$ measure how house prices, rents, and the mortgage rate respectively respond to disequilibrium in the long-run relation. The constant is unrestricted, so the long-run relation may include a non-zero intercept. We estimate (3) by Gaussian maximum likelihood, which requires (i) correct lag length, (ii) i.i.d. Gaussian innovations, and (iii) constant parameters; rank inference is robust to non-normality, but inference on individual coefficients relies on the latter two assumptions.

Rank determination. If $r = 0$ there is no co-integration and the model reduces to a VAR in differences; if $0 < r < p$ there are r stationary long-run relations; if $r = p$ the system is stationary in levels. We test $H_0 : \text{rank}(\Pi) \leq r_0$ against $H_1 : \text{rank}(\Pi) > r_0$ by the Johansen trace statistic

$$\text{LR}_{\text{trace}}(r_0) = -T \sum_{i=r_0+1}^p \ln(1 - \hat{\lambda}_i),$$

where $\hat{\lambda}_i$ are the ordered eigenvalues from the reduced-rank regression. The trace statistic has a non-standard asymptotic distribution depending on the deterministic specification; we use the constant-in-CI / constant-outside-CI variant compatible with an unrestricted constant and the Johansen (1995) critical values. The procedure is sequential, starting at $r_0 = 0$ and stopping at the first non-rejection.

Restrictions on β . A linear restriction of the form $\beta = H\phi$ (equivalently $R'\beta = 0$ for $R = H_\perp$) is tested by the LR statistic

$$\text{LR} = -T \sum_{i=1}^r \left[\ln(1 - \tilde{\lambda}_i) - \ln(1 - \hat{\lambda}_i) \right] \xrightarrow{d} \chi^2(s),$$

where $\tilde{\lambda}_i$ are the restricted eigenvalues and s is the number of restrictions. We use this to test the strict user-cost restriction $\beta_r = 1$, in which case $H = ((1, -1, 0)', (0, 0, 1)')$ and $s = 1$.

4 Empirical Analysis

4.1 Integration Properties

Table 2 reports ADF tests. The level series are clearly non-stationary: the null of a unit root is not rejected for h_t , r_t , or m_t . In first differences the null is rejected for h_t and m_t at the 5% level, while Δr_t is borderline ($p = 0.087$), which is not unusual in a short monthly sample with strong persistence in rents. Taken together, the evidence is consistent with treating the three series as $I(1)$ and proceeding to co-integration testing.

Table 2: ADF unit-root tests with constant; lag length chosen by AIC.

Variable	Stat. (lvl)	p (lvl)	Lags (lvl)	Stat. (Δ)	p (Δ)	Lags (Δ)
h_t	0.2679	0.9758	14	-3.2160	0.0191	1
r_t	1.7251	0.9982	12	-2.6296	0.0870	8
m_t	-1.6255	0.4698	5	-3.6555	0.0048	4

4.2 Lag Selection

Lag length is chosen by estimating a VAR in levels and minimising AIC; Table 3 shows that AIC is minimised at lag 4, with BIC and HQIC also pointing to lag 4 within the displayed range. The preferred VECM therefore uses $k_\Delta = 3$ lagged differences.

Table 3: Lag-order selection criteria for VAR in levels (2012M1–2023M12).

Lag	AIC	BIC	HQIC
0	-10.3391	-10.2736	-10.3124
1	-29.0146	-28.7525	-28.9081
2	-30.7035	-30.2449	-30.5171
3	-30.9077	-30.2525	-30.6414
4	-31.1498	-30.2980	-30.8037
5	-31.1138	-30.0655	-30.6878
6	-31.0758	-29.8309	-30.5699
7	-31.0536	-29.6122	-30.4679
8	-31.0042	-29.3662	-30.3386
9	-31.0864	-29.2519	-30.3409
10	-31.0830	-29.0519	-30.2577
11	-31.0071	-28.7795	-30.1019
12	-30.9735	-28.5494	-29.9885

4.3 Co-Integration Rank

With the dynamic specification fixed, we test the co-integration rank. Table 4 reports the Johansen trace statistic at each r_0 . For $H_0 : r = 0$ against $H_1 : r \geq 1$, the statistic of 48.23 exceeds the 5% critical value of 29.80, so we reject. For $H_0 : r \leq 1$ against $H_1 : r \geq 2$, the statistic of 11.56 is below the 5% critical value of 15.49, so we do not reject. The preferred rank is therefore $r = 1$.

Table 4: Johansen trace test. Sequential procedure rejects if trace stat. exceeds the 5% critical value.

H_0	Trace stat.	CV 90%	CV 95%	CV 99%
$r = 0$	48.2266	27.0669	29.7961	35.4628
$r \leq 1$	11.5648	13.4294	15.4943	19.9349
$r \leq 2$	3.6207	2.7055	3.8415	6.6349

Having found rank $r = 1$, we estimate the CVAR and characterise the long-run relation. Figure 2 plots the actual h_t together with the fitted long-run component implied by the normalised co-integrating vector, and Table 5 reports the implied coefficients with ML standard errors.

Table 5: Preferred CVAR (k=4, unrestricted constant, rank 1). Normalised on h_t . Std. err. from VECM ML.

Parameter	Estimate	Std. err.
β_r (rent elasticity)	1.4918	0.0467
β_m (mortgage coef.)	0.0085	0.0068
α_h	0.0156	0.0049
α_r	0.0091	0.0018
α_m	0.3958	0.5380

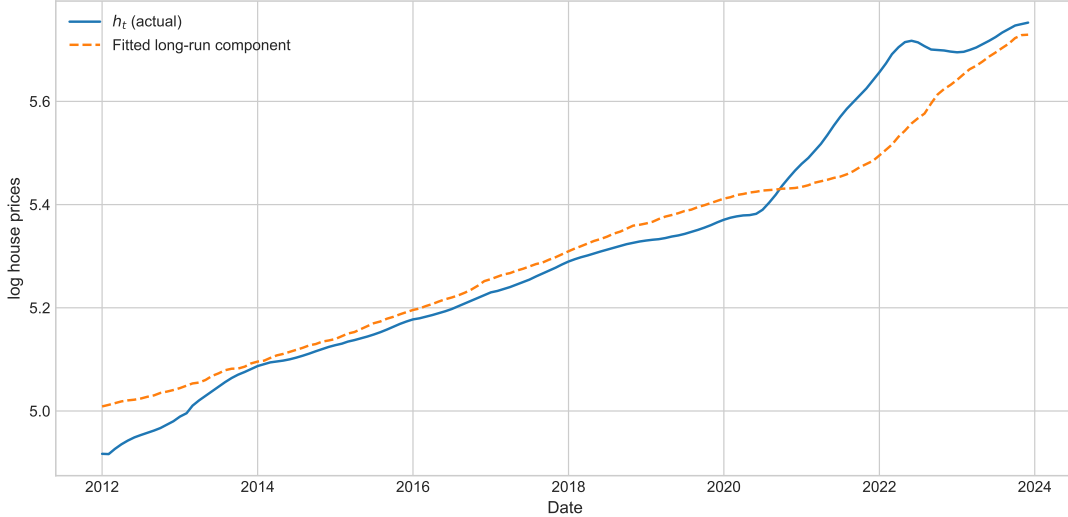


Figure 2: Actual h_t and fitted long-run component from the preferred CVAR.

4.4 Preferred CVAR and the User-Cost Restriction

Normalising on house prices, the estimated long-run relation is

$$h_t = \hat{c} + 1.4918 r_t + 0.0085 m_t, \quad (4)$$

with ML standard errors of 0.0467 on the rent coefficient and 0.0068 on the mortgage-rate coefficient. The rent elasticity is statistically and economically larger than one (a z -ratio of $(1.4918 - 1)/0.0467 \approx 10.5$ rejects $\beta_r = 1$ at any conventional level), and the mortgage coefficient is small, statistically insignificant ($t \approx 1.25$), and has the wrong sign relative to the user-cost prediction $\beta_2 < 0$. The formal Johansen LR test of the strict user-cost restriction $\beta_r = 1$ in Table 6 corroborates this: the LR statistic of 14.46 has a $\chi^2(1)$ p -value of 0.0001, so we strongly reject unit rent elasticity. The data support a stable long-run link between house prices, rents, and the mortgage rate, but they reject the exact sign and magnitude pattern of the textbook model.

Table 6: Johansen LR test of the strict user-cost restriction $\beta_r = 1$. H_0 : unit rent elasticity; asymptotic distribution $\chi^2(1)$.

Test	Statistic	d.f.	p-value
LR test $\beta_r = 1$	14.4616	1	0.0001

The adjustment loadings indicate how equilibrium is restored. The mortgage equation carries almost all of the short-run correction ($\hat{\alpha}_m = 0.3958$, though imprecisely estimated), while $\hat{\alpha}_h = 0.0156$ and $\hat{\alpha}_r = 0.0091$ are statistically different from zero but economically tiny. This is the opposite of the user-cost story, in which house prices adjust to the rent-cost wedge.

4.5 Misspecification Diagnostics

Residual diagnostics in Table 7 are mixed. The Portmanteau whiteness test ($p = 0.30$) finds no remaining serial correlation, so the dynamic specification is statistically adequate in that dimension. The Jarque–Bera test strongly rejects normality, and ARCH-LM rejects homoskedasticity for the mortgage-rate residuals at the 5% level ($p = 0.022$) and is borderline for h_t ($p = 0.086$). These results support consistency of the rank estimate (which does not rely on Gaussianity), but the violations of normality and homoskedasticity imply that the standard errors on β and α should be read with caution and that asymptotic-normality based tests are at best approximate; the post-2021 interest-rate shock is the likely driver.

Table 7: Misspecification diagnostics for the preferred CVAR specification.

Test	Statistic	p-value
Whiteness (Portmanteau)	83.9923	0.3012
Normality (Jarque–Bera)	391.7401	0.0000
ARCH LM (h)	19.1037	0.0861
ARCH LM (r)	6.4636	0.8909
ARCH LM (m)	23.7053	0.0223

5 Discussion

The empirical evidence supports one co-integrating relation between house prices, rents, and the mortgage rate in 2012M1–2023M12, but the economic content of the relation is only partly aligned with the textbook user-cost model: the rent elasticity is significantly above one, the mortgage-rate coefficient has the wrong sign and is insignificant, and the formal LR test of $\beta_r = 1$ is rejected at $p = 0.0001$. Simpler frictions absent from (3) — risk premia, refinancing constraints, supply rigidities, expectational shifts — offer natural explanations for the elasticity exceeding one. Equilibrium is also restored in a way that the textbook does not predict: the mortgage rate, not house prices, does most of the short-run adjusting toward the long-run wedge.

Table 8: Robustness of the main results to extending the sample to 2000M1–2023M12.

Quantity	2012M1–2023M12	2000M1–2023M12
VAR lag (AIC)	4	4
Trace stat. at $r = 0$	48.23	19.64
Trace stat. at $r \leq 1$	11.56	3.68
β_r	1.4918	1.5521
β_m	0.0085	0.1719
LR stat. ($\beta_r = 1$)	14.4616	2.7022
LR p -value	0.0001	0.1002

To gauge robustness we re-estimate the same specification on the longer sample 2000M1–2023M12, which includes the housing boom and the financial crisis. AIC again selects four lags, and the implied long-run coefficients ($\hat{\beta}_r = 1.55$, $\hat{\beta}_m = 0.17$) are qualitatively similar. The trace statistic at $r = 0$ collapses to 19.6, however — below the 5% critical value of 29.8 — so we cannot even reject no co-integration on the longer sample, and the LR test of $\beta_r = 1$ has $p = 0.10$, no longer significant at the 5% level. The 2012–2023 finding is therefore sample-specific: the bubble–bust period is informative enough about the data-generating process that the post-crisis “stable equilibrium” interpretation does not survive its inclusion. This sensitivity, combined with the residual diagnostics, suggests the estimated long-run relation should be read as a useful approximation for the post-crisis regime rather than a structural law.

6 Conclusion

For 2012M1–2023M12 a CVAR with $k = 4$ lags identifies one co-integrating relation among house prices, rents, and the mortgage rate. The strict user-cost benchmark is rejected: $\hat{\beta}_r = 1.49$ with $p < 10^{-4}$ for $H_0 : \beta_r = 1$, and the mortgage-rate coefficient is small, insignificant, and of the wrong sign. Equilibrium is restored mainly through the mortgage-rate equation, and the finding does not survive extension of the sample to include the 2000s boom and the 2008-09 crisis.

References

- Anundsen, A.K. (2019). Detecting Imbalances in House Prices: What Goes Up Must Come Down? *The Scandinavian Journal of Economics*, 121(4), 1587–1619.
- Johansen, S. (1995). *Likelihood-Based Inference in Cointegrated Vector Autoregressive Models*. Oxford University Press.